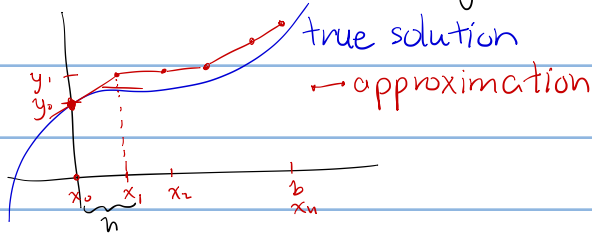


Numerical Solutions of Differential Solutions

Limit definition of derivative

$$\lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$



$$\frac{y_{n+1} - y_n}{h} = f(x_n, y_n) \quad y_{n+1} = y_n + h f(x_n, y_n)$$
$$x_n = x_0 + nh$$

approximate with secant

$$\frac{dy}{dx} = \overbrace{2xy}^{f(x,y)}, \quad x_0=0, y_0=1, \quad y_0(0)=1 \quad [0, 2] \quad h = \frac{1}{2}$$

$$x_1 = \frac{1}{2}, \quad x_2 = 1, \quad x_3 = \frac{3}{2}, \quad x_4 = 2$$

$$y_1 = y_0 + h f(x_0, y_0) \quad y_2 = y_1 + h f(x_1, y_1) \quad y_3 = y_2 + h f(x_2, y_2)$$
$$= 1 + \frac{1}{2} 2(0)(1) \quad = 1 + \frac{1}{2} 2 \cdot \frac{1}{2} \cdot 1$$

$$y_1 = 1 \quad = 1 + \frac{1}{2} \quad y_4 = y_3 + h f(x_3, y_3)$$
$$= \frac{3}{2}$$

Forward Euler Method